

Aadhaar Bhatia

B.E Mathematics & Scientific Computing · BITS Pilani · Expected 2028
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EDUCATION

B.E in Mathematics and Scientific Computing

Birla Institute of Technology and Science (BITS) Pilani
Minor in Finance

Graduating May 2028

CGPA: 8.57 / 10

Class XII — Montfort Sr. Sec. School, New Delhi

95.5% (2024)

Class X — Montfort Sr. Sec. School, New Delhi

95.4% (2022)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL

AI / ML: CrewAI, LLM Integration (GPT-4o, open-source LLMs), Pydantic, Prompt Engineering, Streamlit, NumPy, Matplotlib, MATLAB, scikit-learn · Linear & Logistic Regression implemented from scratch in NumPy

Web & Full-Stack: React, Firebase (Firestore, Auth, Storage), Algolia, REST APIs, FastAPI

Data & Scraping: Pandas, BeautifulSoup, Requests, PyMuPDF, SQLite, Altair

Competitive Programming: 100+ LeetCode problems solved (Arrays, Strings, Linked Lists, Trees, Graphs, Dynamic Programming)

EXPERIENCE

Software & AI Engineering Intern | Samaro AI

May 2026 – Present

- Reverse-engineered a Next.js production application using Chrome DevTools to identify its underlying Supabase REST API — eliminating browser automation entirely and building a Python pipeline that programmatically queries the API with dynamic search parameters across locations and photography categories.
- Processed JSON responses via Pandas with automated data cleaning, deduplication, and normalization into structured tabular datasets; exported to both CSV and SQLite — building a scalable photographer database across India designed for thousands of records.
- Architecting and deploying agentic AI chatbots for automated voice calls and consulting workflows, and building a smart CRM system from the ground up covering data modelling, API design, and front-end integration; evaluating open-source LLMs for production use-cases, benchmarking latency, accuracy, and cost trade-offs.

PROJECTS

HireWise — AI Resume Screener [Python · CrewAI · GPT-4o · Pydantic · PyMuPDF · Streamlit]

- Architected a 4-agent sequential CrewAI pipeline (JD Analyst → Resume Screener → Deep Analyst → Report Writer) with explicit context chaining — each agent receives cumulative outputs from all prior agents, eliminating redundant re-processing and preserving decision continuity across the workflow.
- Enforced output integrity via Pydantic schemas with field-level constraints (e.g. skill weights validated $ge=1$, $le=5$; scores clamped 0–100) — malformed agent outputs are rejected at the schema layer before propagating downstream; instantiated a fresh crew per candidate to prevent inter-candidate memory bleed.
- Built two custom CrewAI tools from scratch — a PyMuPDF-based PDF extractor with page-by-page parsing and graceful fallback for scanned/image-only PDFs, and a directory scanner returning structured JSON — deployed via Streamlit UI with a configurable pass/fail threshold and auto-saved per-candidate markdown reports.

ReFIND — Student Second-Hand Marketplace [React · Firebase (Auth, Firestore, Storage) · Algolia · Vercel]

- Engineered an atomic 'mark as sold' flow as a single Firestore writeBatch committing 7 document operations simultaneously — updating product stock, writing to seller's productsSold and buyer's productsBought subcollections, dispatching a buyer notification, and incrementing both users' wallet balances and CO2 totals via increment() — preventing any partial-state race conditions.
- Built a custom useChat hook resolving Firestore's lack of native OR queries by running two parallel onSnapshot listeners (buyerId / sellerId), merging and deduplicating results via Map keyed on chat ID; initiateChat checks for an existing buyer-seller-product triple before creating a new room, preventing duplicate conversations.
- Integrated Algolia InstantSearch with faceted RefinementList, live SearchBox, and pagination synced to Firestore; on every new listing, a dynamically chained compound query (category → subCategory → size) cross-references buyer requests and dispatches batched match-notifications in a single commit, deduplicating recipients via Set.

IEX Electricity Market Analysis Pipeline *[Python · Streamlit · Pandas · Altair · Matplotlib]*

- ▶ Built a 6-tool Streamlit pipeline for Indian Energy Exchange market analysis — automated header/aggregate-row detection, DAM-RTM merging, multi-file consolidation, bid reconciliation, and interactive plot generation — replacing a fragmented manual Excel workflow with a single-click system across 10 years of Day Ahead Market data (96 time blocks/day).
- ▶ Engineered a configurable Dynamic Price Cap Simulator with a stateful day-by-day cap adjustment algorithm: consecutive-day trigger window, percentage-based step size, and post-adjustment lock period to prevent oscillation; outputs downloadable daily/monthly CSVs and PNG charts.
- ▶ Developed a Buy/Sell Bid Reconciliation tool cross-validating CERC regulatory files against IEX website data at 3 decimal place (MU) precision; findings presented to VP and Technical Head and adopted for ongoing internal market design analysis.

LEADERSHIP & EXTRACURRICULARS

Consultant — 180 Degrees Consulting (180DC)

2025 – Present

- ▶ Advising early-stage startups on business challenges and growth strategy using data-driven problem solving.

Training & Counselling Lead — Student Welfare & Mentorship Committee (SWMC)

- ▶ Recruiting, training, and onboarding a cohort of over 250 student mentors, creating structured peer-counselling programmes adopted campus-wide.

Student Coordinator — Student Alumni Relations Cell (SARC)

- ▶ Managing vendor relations, alumni hosting logistics, and GB communications for AY 2026.

Volunteer — National Centre for Disease Control (NCDC)

June 2025

- ▶ Supported workshop coordination for public health events, assisting with marketing logistics, design materials, and on-ground operations.

Competitions & Other: Break The Case 2025 (case competition participant); regular MUN & debate conference participant; campus journalism and public speaking societies.